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A micro-influencer recommender method based on brand features using deep learning approaches

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ABSTRACT

Influencer marketing has become a critical strategy for brands, with micro-influencers playing a pivotal role in shaping consumer behavior on social media. These influencers significantly impact user decision-making; however, identifying the right micro-influencers who align with a brand's identity and can maximize engagement remains a challenging task. This study aims to develop an effective system to predict which micro-influencers will generate the highest engagement for brands. The proposed method combines ResNet for extracting visual features from Instagram posts and BERT for extracting textual features, demonstrating a high degree of accuracy in predicting engagement rates. Three machine learning algorithms—Linear Regression, Random Forest, and Extreme Gradient Boosting—were employed for model training, with performance evaluated using Root Mean Squared Error (RMSE). Among these, the Random Forest model, enhanced through feature selection and hyperparameter tuning, achieved the lowest error with an RMSE of 1.50 %. Further evaluation using metrics such as AUC, cAUC, and MedR demonstrated that the proposed approach (ResNet + BERT) outperforms other methods across all metrics, achieving an AUC of 81 %, a cAUC of 67 %, and a MedR of 2. The proposed method improved AUC by 12 % and cAUC by 8 %, confirming its effectiveness in identifying high-engagement micro-influencers.

1. Introduction

In recent years, influencer marketing has become a crucial element of social media marketing, and brands have recognized its importance [1]. Customers perceive influencer opinions about products as more credible [2,3]. Micro-influencer marketing on social media platforms has become an essential advertising strategy for brands seeking to leverage influencers with smaller but highly engaged audiences. However, identifying suitable micro-influencers that align with a brand's image and style can be challenging due to the large number of potential candidates [4–6]. There is no open-access database that connects brands to suitable micro-influencers, which brands could utilize directly. Brands must manually search social media platforms to find relevant micro-influencers [6–8]. Additionally, determining which micro-influencer is more suitable for a brand requires expertise in evaluating factors such as audience demographics, content style, engagement metrics, etc., which most brands lack. Furthermore, this process is time-consuming and costly for brands.

While previous studies has explored methods for predicting influencer engagement, many approaches rely on basic techniques for content analysis. Several studies have examined various aspects of influencer marketing, particularly focusing on micro-influencers and their impact. Studies [6,8,9] have developed methods for ranking micro-influencers based on factors such as brand fit, content analysis, and future engagement prediction. They utilized machine learning techniques and social media data to identify the most effective micro-influencers for brand promotion. Research in [10,11] investigated the impact of visual elements such as image alignment on consumer-brand engagement, highlighting the importance of visual alignment between influencer content and the brand. In [12], the use of machine learning algorithms for identifying social media micro-influencers was explored, demonstrating the potential of data-driven approaches in this area.

Previous studies have employed traditional machine learning for extracting visual and textual features, neglecting the more advanced models developed recently. In this research, we aim to predict micro-influencer engagement by developing a more sophisticated method. We propose a novel approach that leverages the power of deep learning

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Table 1
Comparison of related micro-influencer recommendation systems.

Ref	Year	Methodology	Dataset	Advantages	Disadvantages
[6]	2019	Pre-trained CNN, Word2Vec, MIR ranking model	Instagram dataset of 360 brands and 3748 micro-influencers	Provides a rich and accurate understanding of micro-influencers' perspectives and experiences.	small sample size of participants Reliance on self-reported data in interviews and surveys.
[9]	2022	ResNet, UperNet, MORNING ranking	Instagram dataset of 360 brands, 3748 micro-influencers, and active audiences	Use multifaceted audience information for learning audience representations	Rely on crawling additional audience information
[4]	2021	VGG16, Tok2Vec	Instagram dataset of 360 brands and 3748 micro-influencers	Uses multi-task learning, Interpretable framework	Relies on the availability and quality of data. Proposed framework is costly.
[11]	2019	ResNet50, FastText, TF-IDF for ranking	Collected images and tags of 1000 followers of each brand and their recent posts on Instagram	Requires less manual effort than questionnaires to measure brand similarity.	Requires a larger number of followers for each brand. Brand similarity alone may not capture complex user preferences and purchasing behavior.
[10]	2020	VGG19, ResNet50, InceptionV3	Collected over 45,000 images and Instagram usage behaviors over 26 months	Automatic image classification	Deep learning algorithms rely on the quality and diversity of training datasets.
[8]	2021	Concept-based ranking, CAMERA, loss function for learning ranking model	Instagram dataset including 360 brands and 3748 micro-influencers and its expansion	Ability to provide a more targeted approach for brands to micro-influencer marketing.	Relies on data mining and social network analysis.
[12]	2022	Random Forest, Decision Tree, Linear Regression	Twitter dataset with 178,819 records	Ability to provide more accurate and comprehensive influencer profiles	Effectiveness of the method depends on the availability and quality of data.
[23]	2018	VGG16	Over 1.1 million image and video posts from 927 brands in 14 Instagram categories	Provides improved quantitative performance and qualitative insights into discovering brand-related content	Limited quantitative and qualitative analysis in relation to brands
[24]	2021	TF-IDF, Naive Bayes, Logistic Regression	Three datasets including 4.14 million tweets and popular hashtags on Twitter in Pakistan	Allows for multilingual analysis of micro-influencers (English and Urdu)	Binary classification of hashtags into political and non-political categories may not be generalizable to other countries

to extract richer features from the visual and textual elements of micro-influencer posts. Therefore, the aim of this research is to propose a micro-influencer recommendation method based on brand characteristics using deep learning, which utilizes a combined approach of ResNet[1] and BERT[2] to enhance accuracy and efficiency in predicting engagement rates.

In summary, the contributions of our research are as follows:

- By combining ResNet for visual feature extraction and BERT for textual feature extraction, we introduce a novel approach to comprehensively represent micro-influencer posts, capturing both visual and textual cues that influence engagement.
- Through rigorous experiments with various machine learning algorithms, we develop a metric for predicting micro-influencer engagement that demonstrates the superior performance of Random Forest with feature selection and hyperparameter tuning.
- With a deeper understanding of micro-influencer posts, the proposed method can recommend highly engaging content with greater consistency across the ranked list.

2. Related work

This section will delve into a review of prior research in the domain of machine learning.

2.1. Micro-influencer marketing

Numerous studies have examined various aspects of influencer marketing, with a particular focus on micro-influencers and their impact. Research [6,8,9] has explored methods for ranking micro-influencers based on factors such as brand fit, content analysis, and future engagement prediction. These studies have employed machine learning techniques and social media data to identify the most effective micro-influencers for brand promotion. Research [10,11] has investigated the influence of visual elements, such as image alignment, on consumer-brand engagement, highlighting the importance of visual consistency between influencer content and the brand. In [12], the use of machine learning algorithms for identifying social media influencers was explored, demonstrating the potential of data-driven approaches in this area. Overall, these similar works underscore the importance of data and machine learning in optimizing influencer marketing strategies, particularly in the realm of micro-influencer selection.

2.2. Learning to rank

Learning to Rank (LtR) is an active research area in machine learning [13–15]. Generally, LtR techniques aim to rank data instances such as documents or images, using either labeled ranking examples (training data) or considering an active learning scenario where rankings are iteratively generated from implicit or explicit user feedback [16].

Several studies have applied LtR techniques to address the challenge of selecting influencers for brand marketing campaigns. Some studies [4,6,8,9] implemented various ranking algorithms to prioritize influencers based on their predicted engagement. These studies utilized machine learning models trained on social media data, including follower demographics, engagement metrics, content analysis, and historical performance. By learning from this data, the models predict each influencer's potential impact on the brand. Ranking algorithms then rank influencers according to these predictions, enabling brands to focus on those with the highest predicted engagement and brand alignment. In [6], a listwise Multiple Influencer Ranking (MIR) model was proposed to predict ranking scores for brands and micro-influencers. Similarly, [9] introduced a joint learning framework for micro-influencer ranking, referred to as MORNING. In [4], a ranking method based on multi-task learning and an interpretable framework was proposed, combining multiple tasks to simultaneously learn and rank influencers based on their impact on social media. Additionally, [8] proposed a data-driven approach called CAMERA, a concept-based micro-influencer ranking framework consisting of three components: social media concept learning, social media account representation learning, and micro-influencer ranking.

[1] Residual Neural Network
[2] Bidirectional Encoder Representations from Transformers

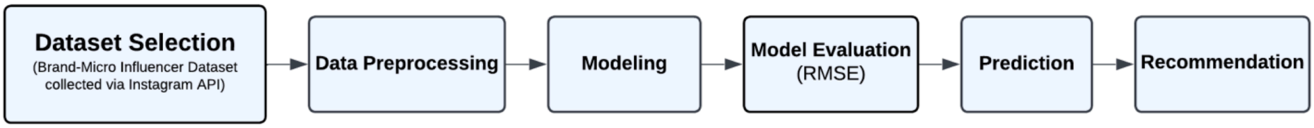

Fig. 1. An overview of the proposed method.

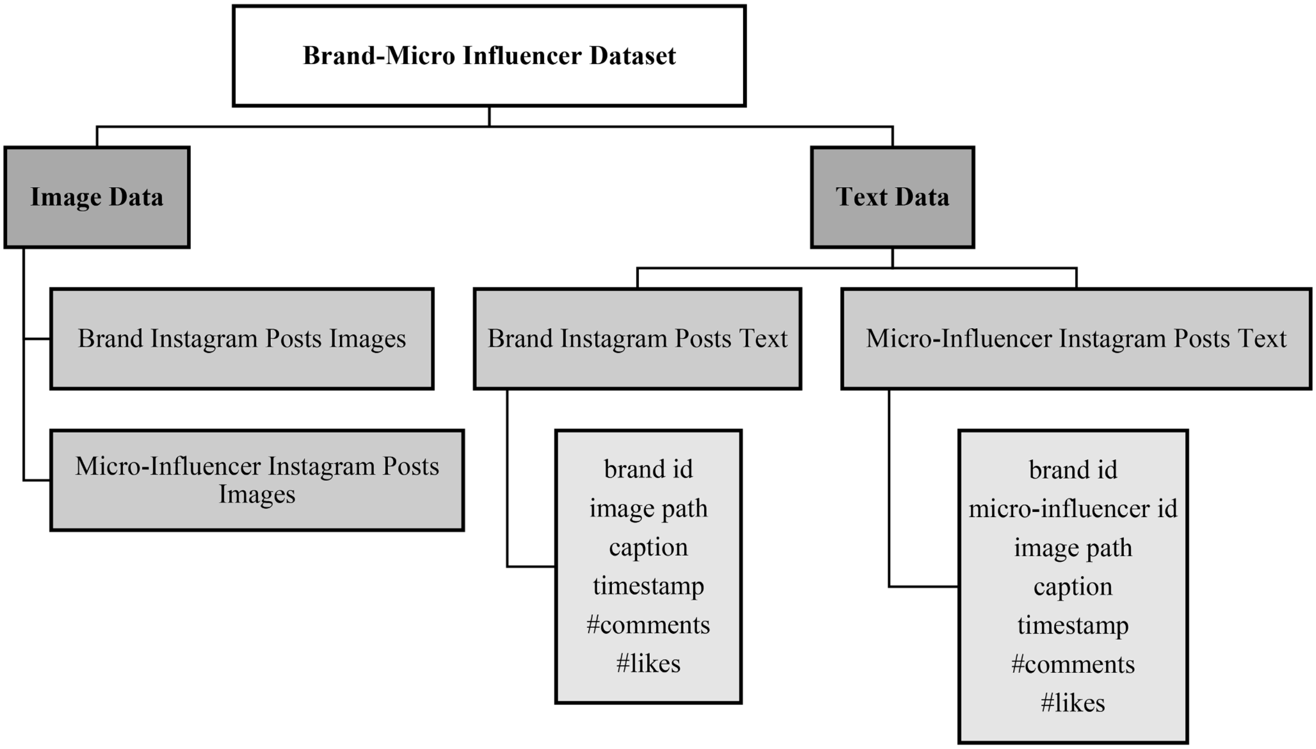

Fig. 2. Overview of the Brand-Micro Influencer Dataset Structure.

2.3. Recommender system

A recommender system is a technology that helps us discover things we might like. A recommender system can analyze past purchases, browsing history, and even things like what other people with similar tastes have liked [17–20]. Based on this information, it suggests products that we are most likely to be interested in. These systems are used everywhere from online stores to social media platforms and act like a personal assistant [21,22]. As seen in related works [4,6,8,9], various applications of recommender systems in this area have been investigated. These studies considered influencers as "items" to be recommended, with the goal of ranking them. In [9], micro-influencers were recommended to brands through their joint learning framework. Table 1 shows a comparison of the related works.

3. Methodology

Representing brands and micro-influencers is the foundation for effectively recommending them to brands. However, there is no standardized method for such representation. Therefore, we proposed a novel multimodal social account embedding method that leverages historical social media posts to represent brands and micro-influencers. After generating the multimodal social account representation, our proposed hybrid model was developed using ResNet for visual feature extraction and BERT for textual feature extraction. The combined model was trained with the dataset and utilized to predict engagement scores between brands and micro-influencers. Finally, the predicted engagement scores were employed to rank the micro-influencers, and the ranking was used to provide optimized recommendations for brand promotion. The proposed framework is illustrated in Fig. 1.

3.1. Dataset characterization

In this research, we utilized the open-source Brand-Micro Influencer Dataset introduced in [6]. This dataset was constructed by crawling publicly available Instagram data using the platform's official API. It encompasses data from 360 brands across 12 industry sectors[3] and 3748 micro-influencers. Each entity corresponds to a specific Instagram account.

For the data collection process, up to 50 of the most recent posts of each brand and micro-influencer were retrieved, including visual content (post images), textual content (captions), and engagement metrics such as the number of likes and comments, alongside profile data like follower count and biography information.

The data collection process for Instagram micro-influencers involved three stages:

Stage 1. A list of 1000 top micro-influencers, based on the highest follower count, was compiled from the 3748 micro-influencers mentioned in source [6].

Stage 2. For each micro-influencer, links to their Instagram posts were extracted.

[3] Airlines, Auto, Fashion, Food, Furnishing, Electron., Jewellery, Finance, Services, Entertain., Energy, and Beverages.

Table 2
Performance comparison across different modeling scenarios.

Model	Feature selection	Tuning hyperparameters	RMSE (train)	RMSE (test)	RMSE (all)
Linear Regression	No	No	2.012	2.022	2.015
	No	Yes	–	–	–
	Yes	No	2.052	1.999	2.039
	Yes	Yes	–	–	–
Random Forest	No	No	0.730	1.922	1.150
	No	Yes	1.680	1.693	1.683
	Yes	No	0.729	1.953	1.163
	Yes	Yes	1.509	1.501	1.507
XGBoost	No	No	1.498	1.882	1.603
	No	Yes	1.714	1.682	1.706
	Yes	No	1.537	1.944	1.648
	Yes	Yes	1.744	1.737	1.742

Stage 3. Information such as the number of likes, comments, captions, etc. was retrieved from each post.

The structure of the Brand-Micro Influencer dataset is visualized in Fig. 2. This dataset consists of two main components: brands and micro-influencers, each associated with a rich set of attributes that capture their content and engagement behavior on Instagram. The dataset is organized into textual and visual folders, where each brand and influencer has a dedicated set of Instagram posts.

For each entity (brand or micro-influencer), the dataset includes metadata for every post, such as caption_text, which holds the descriptive text accompanying the post, and timestamp, which records the exact posting date and time. The fields number of likes and number of comments quantify the audience engagement for each post and serve as core indicators for measuring influence. The fields path1 and path2 refer to the file locations of the two images associated with each post—typically used for visual feature extraction in multimodal analysis.

In addition to post-level data, the dataset includes biographical information contained in brand_biography and influencer_biography, which summarize the profile descriptions. There is also a field named all_positive that identifies confirmed brand–influencer pairs suitable for recommendation tasks. Furthermore, the dataset includes identifiers such as brand_id and micro_influencer_id to uniquely distinguish entities, and the category field categorizes each brand and influencer into one of the 12 defined sectors.

In the influencer section of the dataset, the field brand_id also appears. This is because some posts by micro-influencers are branded collaborations — meaning that they were published on the influencer's personal page but in sponsorship or affiliation with a specific brand. Therefore, the presence of brand_id in both brand and micro-influencer data structures is not redundant, but instead provides a necessary link for modeling brand-influencer relationships.

Subsequently, data preprocessing was conducted, encompassing three stages: data cleaning, feature engineering, and feature selection. The dataset underwent cleaning to remove incomplete data. Visual features were extracted using ResNet50, and textual features were processed with BERT to capture semantic and structural aspects of captions. Engagement metrics like average likes and comments were calculated. The target variable (y_month_01) was defined as the next month's average engagement, and 20 key features were selected for modeling using a correlation heatmap.

3.2. Modeling

Our proposed method is a combination of ResNet [25] and BERT [26, 27] models. We experimented with several scenarios during the modeling process to find the best model (1. without feature selection and without hyperparameter tuning, 2. without feature selection and with hyperparameter tuning, 3. with feature selection and without hyperparameter tuning, 4. with feature selection and with hyperparameter tuning).

In this section, we will discuss the implementation of three machine learning algorithms: linear regression, random forest, and extreme gradient boosting, along with the process of obtaining optimal parameter values for each algorithm. To implement these algorithms, the dataset is divided into training and testing sets. 25 % of the data is used for testing, and the remaining 75 % is used for training. Subsequently, Root Mean Squared Error (RMSE) is calculated as a metric to evaluate the model's performance on the training data, testing data, and the entire dataset.

To extract features, we employed a pre-trained ResNet model, renowned for its exceptional performance in image recognition. By feeding our image data into ResNet, we extract high-level features that transcend basic pixel values [28–30]. These extracted features essentially encapsulate the visual narrative of each post, summarizing the "what" of the image content.

Beyond visual features, we also delved into extracting textual features. Captions, comments, and other textual elements often convey the influencer's message, brand story, or call to action. To extract captions, comments, and other textual elements, we utilized a pre-trained BERT model. These extracted features represent the "why" behind the post, summarizing the textual message and the influencer's intent.

Once we extracted both visual and textual features, we combined them into a single feature vector for each post [31,32]. This vector transforms into a rich and comprehensive representation of the entire influencer post. It encompasses not only the "what" (visual content) but also the "why" (textual content) behind the post.

To improve model performance, we took a step-by-step approach. First, we built a basic model as a starting point. Then, we gradually added complexity by fine-tuning the model's settings (hyperparameter tuning) and selecting the most important features (feature selection) [33,34].

- **Hyperparameter Tuning:** This involves adjusting the model's settings, such as the number of trees in Random Forest, to make it better at recognizing patterns in the data.
- **Feature Selection:** This step identifies the most useful information in the dataset while removing unnecessary or less relevant details, helping the model learn more effectively.

Finally, we combined these two techniques to achieve the best results. This combined method ensures the model uses the most important features and has optimized settings, enabling it to make highly accurate predictions about micro-influencer engagement. By progressively refining the process, our model becomes more effective at uncovering deeper insights and delivering superior performance.

Table 2 summarizes the performance of different modeling scenarios evaluated during the development of the micro-influencer recommendation system. Each scenario combines specific preprocessing techniques, such as feature selection and hyperparameter tuning, with different models, including Linear Regression and Random Forest. The results demonstrate that the combination of feature selection and

Table 3
Comparative performance of Linear Regression, Random Forest, and XGBoost models across various hyperparameter combinations.

Model	Hyperparameter combination	RMSE (Train)	RMSE (Test)	RMSE (All)
Linear Regression	Baseline (No Feature Selection, No Tuning)	2.01	2.02	2.01
Random Forest	n_estimators=100, max_depth=3, max_features='auto'	1.73	1.92	1.15
Random Forest	n_estimators=200, max_depth=5, max_features='sqrt' (Best)	1.50	1.50	1.50
Random Forest	n_estimators=300, max_depth=7, max_features='auto'	1.68	1.69	1.68
XGBoost	n_estimators=100, max_depth=3, learning_rate=0.01	1.74	1.73	1.74
XGBoost	n_estimators=400, max_depth=5, learning_rate=0.01 (Best)	1.50	1.50	1.50
XGBoost	n_estimators=500, max_depth=7, learning_rate=0.1	1.72	1.71	1.71

	username	avg_total_engagement_june	prediction_avg_total_engagement_july	difference_june_july	category
259	hanahaho	1.58	2.03	0.45	Growing
134	btsjungk00kie	2.51	2.58	0.07	Growing
31	alisyakieb	3.42	3.38	-0.04	Declining
447	nussaofficial	0.41	0.61	0.20	Growing
409	monaratuliu	1.39	1.55	0.16	Growing
455	onewithnatalia	20.94	14.03	-6.91	Declining
508	rintiksedu	5.56	6.05	0.49	Growing
278	igpersib	0.89	1.11	0.22	Growing
361	lokernas	0.24	0.60	0.36	Growing
298	iniadzwaaurell	1.83	2.22	0.39	Growing

Fig. 4. Classification results of micro-influencers based on engagement.

hyperparameter tuning with the Random Forest model achieves the lowest RMSE across all datasets, indicating it is the most effective approach for this task.

We compare the performance of three machine learning models (Linear Regression, Random Forest, and XGBoost) in Table 3 on the task of predicting micro-influencer engagement. Each model was evaluated under different hyperparameter configurations and scenarios (with or without feature selection and hyperparameter tuning).

Random Forest achieved the best performance with an RMSE of 1.50, when both feature selection and hyperparameter tuning were applied. The ability of Random Forest to handle feature interactions and its robustness to overfitting make it highly suitable for this task. While XGBoost also achieved an RMSE of 1.50, its computational complexity and time requirements were higher compared to Random Forest, leading to its secondary placement.

3.3. Prediction

The goal of this research is to develop a robust model capable of accurately predicting micro-influencer engagements.

Based on the modeling process and evaluation of four different scenarios, the Random Forest model with feature selection and hyperparameter tuning achieved the best performance with the lowest RMSE values: 1.5 % for training, testing, and overall data.

To make predictions, the selected Random Forest model (referred to as rf_tuning_best_2) was applied to new data. For this purpose, non-essential columns, such as username, year_month, and y_month_01, were excluded from the df_prediction_2 dataset to create the data_predict dataset. These columns do not provide direct input for prediction.

Using the trained model, predictions were made for the target variable, representing the average engagement of influencers in July. The predicted values were added to the original dataset (df_prediction_2) in a new column titled avg_engagemnt_prediction_july. The Random Forest model was also serialized and saved as a file named best_model_rf.pkl for future use.

3.3.1. Engagement trend analysis (Growing or declining)

In addition to predictions, the results were analyzed to categorize influencers as "Growing" or "Declining" based on their engagement trends. The difference between the predicted engagement for July and the actual engagement for June was calculated for each influencer.

- Influencers with a positive difference (predicted engagement > actual engagement) were categorized as "Growing."
- Influencers with a negative difference were categorized as "Declining."

The distribution of influencers in each category is illustrated in Fig. 4, which shows the dominance of the "Growing" category. Specifically, 513 micro-influencers were identified as having increasing engagement (Growing), while 123 influencers experienced declining engagement (Declining). This suggests that most influencers in the dataset exhibit a positive growth trend in July.

3.3.2. Feature importance analysis

To better understand the behavior of the Random Forest model and the key factors influencing engagement predictions, the feature importance scores were analyzed. Feature importance quantifies the contribution of each input variable to the model's predictions.

Fig. 5 illustrates the relative importance of features, where variables related to likes and total interactions demonstrate the highest

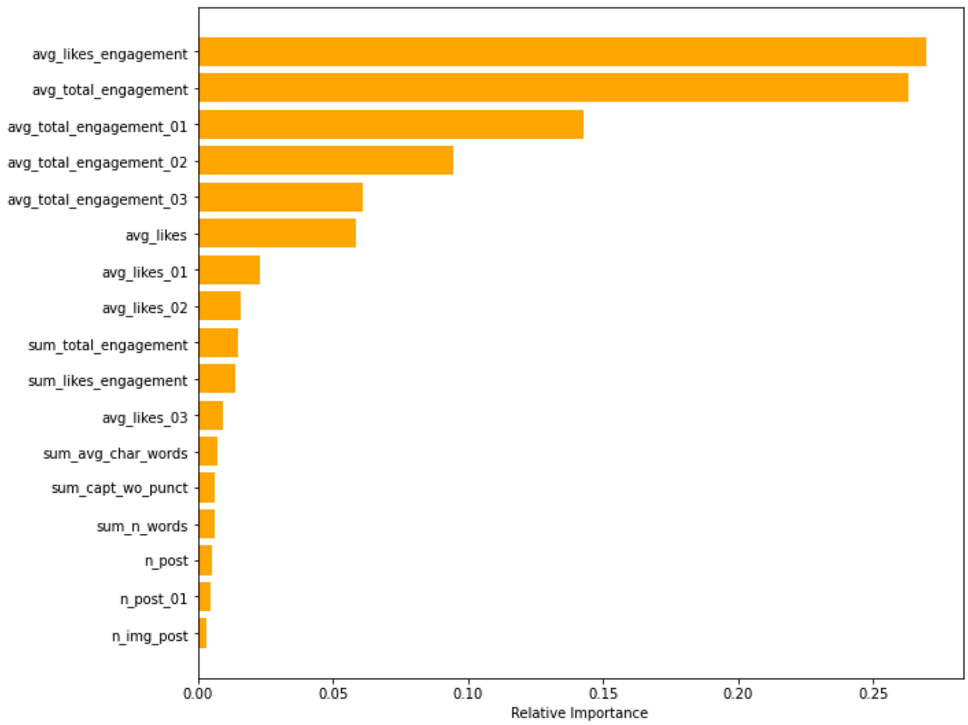

Fig. 5. Bar chart of the relative importance of features related to likes and comments.

importance. These features significantly influenced the model's predictions, highlighting their role as critical indicators of micro-influencer engagement levels.

This analysis reinforces the significance of leveraging interaction-related metrics to predict influencer engagement effectively.

3.4. Recommendation

After predicting engagement rates, the best-performing model is used to generate accurate and tailored recommendations. These recommendations rank micro-influencers based on their predicted engagement potential, helping brands identify those most likely to achieve impactful results.

The system provides a clear framework for selecting influencers suitable for collaboration, supported by key metrics and performance indicators. This allows brands to align influencer partnerships with their marketing goals and target audience. By relying on these data-driven insights, brands can improve the effectiveness of their campaigns and maximize engagement outcomes. This approach ensures informed and efficient decision-making, enabling brands to build successful and high-impact collaborations with top-performing influencers. In the following sections, we will provide a more detailed explanation of this process.

4. Evaluation

In this section, we evaluate the performance of different machine learning models (Linear Regression, Random Forest, and XGBoost) for predicting micro-influencer engagements. We compare these models across various scenarios, including feature selection and hyper-parameter tuning. The evaluation metrics, such as RMSE (Root Mean Squared Error), are used to determine the most accurate model. Furthermore, the section discusses the importance of input features and the trends identified in micro-influencer engagements based on the predictions.

Among all the mentioned scenarios, random forest with feature selection and hyperparameter tuning yielded the lowest RMSE, indicating the best performance. The RMSE for the training set, testing set, and the overall dataset was 1.5, 1.5, and 1.5, respectively, achieving the best results.

To examine the performance of our proposed method across different categories, we conducted an experiment and reported the performance of recommendations based on each category. We introduced the evaluation metrics AUC,[4] cAUC,[5] Precision@k, MRR,[6] MAP,[7] and MedR[8] to assess recommendation performance. The formulas for these metrics are as follows:

- **AUC:** AUC measures the area under the ROC[9] curve, which plots the true positive rate against the false positive rate. A higher AUC value indicates better performance:

$$\boldsymbol{AUC} = \int (\boldsymbol{TPR}(\boldsymbol{FPR}))\ \boldsymbol{dFPR} \tag{1}$$

- **cAUC:** cAUC is similar to AUC but only considers positive and negative samples belonging to a specific category.
- **Precision@k**: Precision@k measures the proportion of relevant items recommended to the user among the top-k recommendations. A higher P@10 value indicates that the model retrieves micro-influencers with high engagement potential among the top-10 results, and similarly for P@50 among the top-50 results:

$$\boldsymbol{Precision@k} = (\boldsymbol{number\ of\ relevant\ items\ in\ top\ k})/\boldsymbol{k} \tag{2}$$

- **MRR:** MRR is the average of the reciprocal ranks of the first relevant item in a ranked list of recommendations. A higher MRR value indicates better performance:

[4] Area Under the Curve
[5] Category-specific Area Under the Curve
[6] Mean Reciprocal Rank
[7] Mean Average Precision
[8] Median Rank
[9] Receiver Operating Characteristic

Table 4
Comparison of the proposed method with related works.

Model	Feature selection	Tuning hyperparameters	RMSE train	RMSE test	RMSE all
CNN + Word2Vec [6]	No	No	2.01 %	2.02 %	2.01 %
	No	Yes	2.11 %	1.99 %	2.01 %
	Yes	No	2.05 %	1.99 %	2.03 %
	Yes	Yes	1.82 %	1.85 %	1.84 %
ResNet + UperNet [9]	No	No	1.91 %	1.99 %	1.91 %
	No	Yes	1.96 %	1.95 %	1.92 %
	Yes	No	2.07 %	1.97 %	1.94 %
	Yes	Yes	1.72 %	1.73 %	1.75 %
VGG-16 + Tok2Vec [4]	No	No	1.91 %	1.99 %	1.91 %
	No	Yes	1.96 %	1.95 %	1.92 %
	Yes	No	2.07 %	1.97 %	1.94 %
	Yes	Yes	1.72 %	1.73 %	1.75 %
ResNet + BERT (proposed)	No	No	0.73 %	1.90 %	1.15 %
	No	Yes	1.67 %	1.69 %	1.68 %
	Yes	No	0.72 %	1.95 %	1.61 %
	Yes	Yes	1.50 %	1.50 %	1.50 %

$$\boldsymbol{MRR} = \sum (1/\boldsymbol{rank}) \Big/ \boldsymbol{total\ number\ of\ queries} \tag{3}$$

- **MAP:** MAP calculates the average precision over all queries. A higher MAP value indicates better performance:

$$\boldsymbol{MAP} = \sum (\boldsymbol{Precision@k} * \boldsymbol{rel}) \Big/ \boldsymbol{total\ number\ of\ queries} \tag{4}$$

- **MedR:** MedR is the median position of the first relevant item in a ranked list of recommendations. A lower MedR indicates better performance.

As shown in Table 4, among all scenarios, the proposed method with feature selection and hyperparameter tuning provides the best results in a timely manner.

Subsequently, we compared the results of implementing our proposed method with [6] for each brand category. The categories of Auto and Clothing exhibited superior performance compared to other categories. Table 5 demonstrates that Auto and Clothing have the highest AUC values (0.915 and 0.873, respectively). Additionally, these two brand categories exhibited better performance in other metrics as well.

We compared the evaluation metric results of the proposed method and [6] for each brand category in a graphical format. As seen in Fig. 6, the AUC metric for the Clothing category in the ResNet + BERT method achieved the highest result among all brand categories with an accuracy of 90 %. Similarly, in the CNN + Word2Vec method, the Makeup category achieved the highest result with an accuracy of 92 %.

In Fig. 7, the cAUC metric for the Clothing category in the ResNet + BERT method achieved the highest result among all brand categories with an accuracy of 82 %, while in the CNN + Word2Vec method, the highest result of 73 % corresponds to the Entertainment category.

As observed, the Auto and Clothing categories exhibited better performance compared to other

Table 5
Performance across brand categories with the proposed BERT and ResNet method.

Category	AUC	cAUC	P@10	P@50	MRR	MAP	MedR	Rec@10	Rec@50
Airline	0.821	0.605	0.234	0.395	0.523	0.224	2	0.393	0.462
Auto	0.873	0.591	0.329	0.545	0.857	0.323	1	0.486	0.593
Clothing	0.915	0.826	0.180	0.487	0.618	0.301	1	0.207	0.512
Drink	0.768	0.697	0.201	0.371	0.589	0.162	1	0.242	0.399
Electronics	0.680	0.641	0.089	0.250	0.282	0.069	9	0.173	0.327
Entertainment	0.803	0.663	0.245	0.419	0.741	0.252	1	0.397	0.501
Food	0.834	0.686	0.256	0.407	0.783	0.276	1	0.285	0.519
Jewelry	0.683	0.607	0.169	0.298	0.467	0.160	2	0.226	0.351
Makeup	0.878	0.734	0.231	0.565	0.806	0.261	1	0.269	0.627
Nonprofit	0.730	0.624	0.116	0.243	0.356	0.106	3	0.184	0.346
Shoes	0.819	0.691	0.254	0.601	0.722	0.262	1	0.283	0.738
Services	0.827	0.725	0.152	0.469	0.451	0.136	2	0.195	0.719

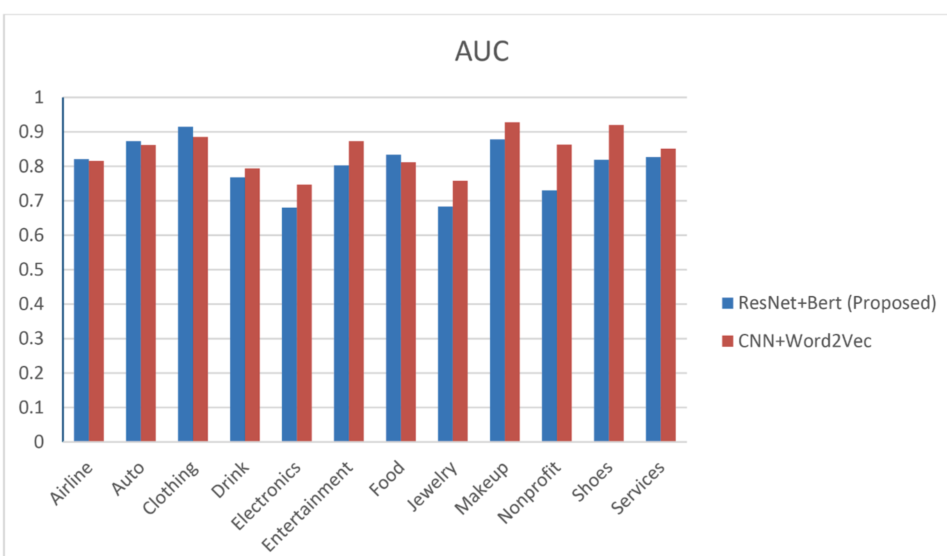

Fig. 6. Bar chart comparing the AUC metric results of the proposed method and related work in brand categories.

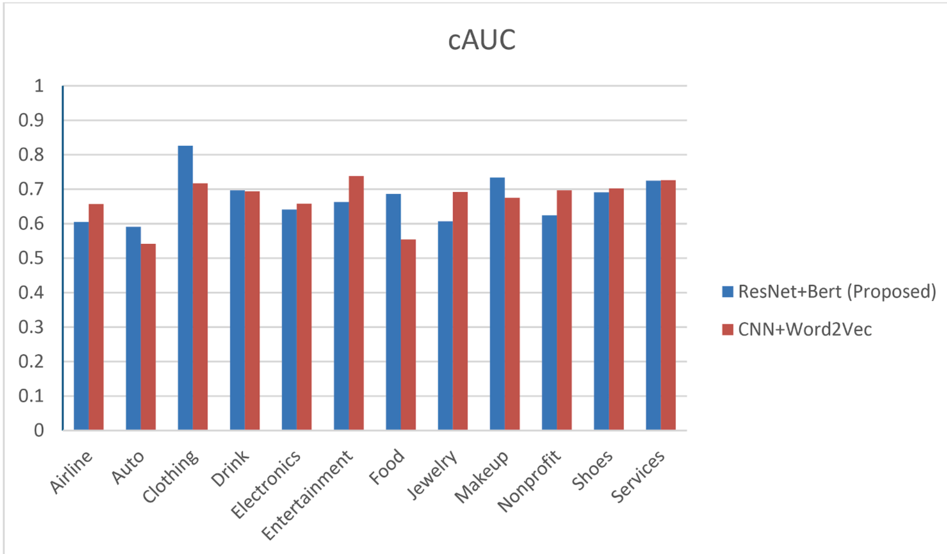

Fig. 7. Bar chart comparing the cAUC metric results of the proposed method and related work in brand categories.

Table 6
Efficiency of the proposed model compared to related works.

Method	AUC	cAUC	P@10	P@50	MRR	MAP	MedR	Rec@10	Rec@50
Pretrained CNN [6]	0.829	0.659	–	–	–	–	8	0.095	0.382
Word2Vec [6]	0.531	0.517	–	–	–	–	57	0.014	0.076
Pretrained CNN + Word2Vec [6]	0.691	0.592	–	–	–	–	26	0.034	0.153
EfficientNet	0.806	0.671	0.124	0.383	0.377	0.161	5	0.249	0.401
BERT	0.789	0.653	0.130	0.350	0.368	0.153	7	0.241	0.483
EfficientNet + BERT	0.808	0.665	0.144	0.420	0.372	0.165	6	0.218	0.522
ResNet	0.779	0.635	0.073	0.313	0.259	0.098	8	0.147	0.408
Proposed (BERT + ResNet)	0.810	0.678	0.207	0.422	0.602	0.213	2	0.347	0.513

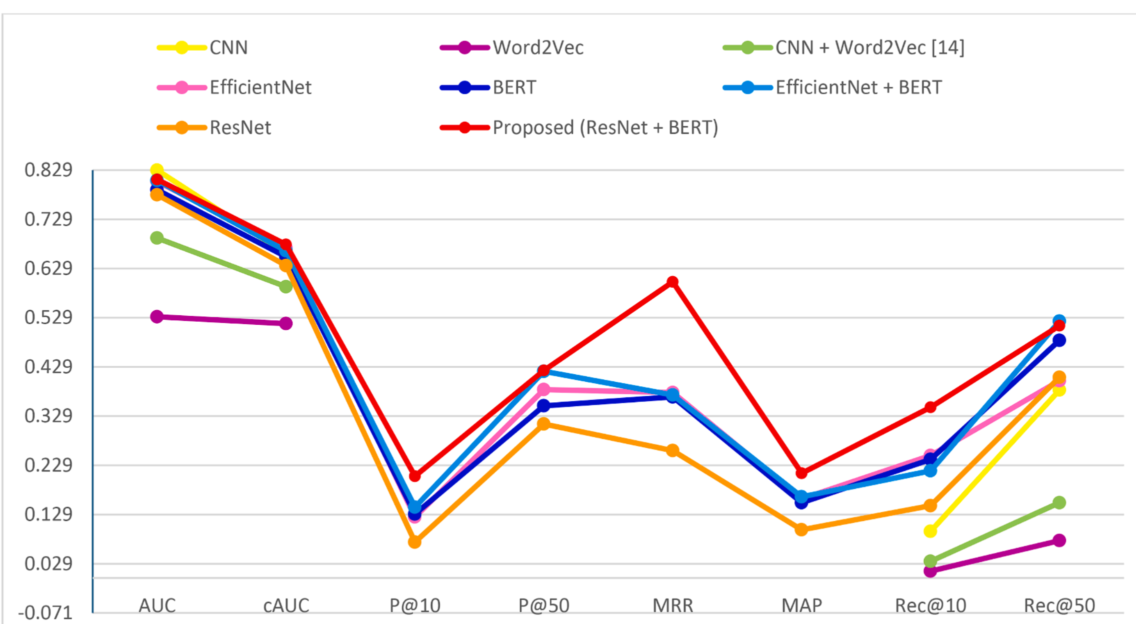

Fig. 8. Comparison of the proposed model's performance with related works.

4.1. Comparison with related works

We compared the evaluation performance results of the proposed method based on the combination of BERT and ResNet with EfficientNet, BERT, the combination of EfficientNet and BERT, and the ResNet model alone. The experimental results in Table 6 show that our proposed method (ResNet + BERT) outperforms other methods in all metrics.

The proposed method (ResNet + BERT) achieves significantly better results in almost all metrics. This indicates that combining visual features extracted by ResNet alongside textual features from BERT provides a more comprehensive understanding of influencer posts, leading to more accurate engagement predictions.

We implemented the proposed method, which combines BERT and ResNet, along with pre-trained CNNs, Word2Vec, CNN + Word2Vec, EfficientNet, BERT, EfficientNet + BERT, and ResNet using the same dataset. Fig. 8 compares the performance of the proposed model with related works. As shown, our proposed method, represented in red, outperforms other methods in all evaluation metrics.

5. Conclusions

This research aims to enhance influencer marketing by developing a model capable of accurately predicting influencer engagement. We employed a hybrid approach, combining ResNet for extracting visual features and BERT for text analysis, to create a comprehensive representation of micro-influencer posts, which demonstrates a high degree of accuracy in predicting engagement rates. Four scenarios were considered, each implemented and executed using Linear Regression, Random Forest, and Extreme Gradient Boosting algorithms, and evaluated based on Root Mean Squared Error (RMSE). Among all scenarios, the proposed method with the Random Forest algorithm, feature selection, hyperparameter tuning, and RMSE=1.50 % yielded the best results. Finally, the obtained results were evaluated using metrics such as AUC, cAUC, Precision@k, MRR, MAP, and MedR, demonstrating that our proposed method outperforms the related work method.

The limitation of this research is that it focused on the Instagram platform. Additionally, the study primarily concentrates on predicting engagement at the post level, without addressing the long-term impact of micro-influencer relationships on brand loyalty and customer behavior.

While this study primarily focuses on Instagram, the proposed framework is inherently flexible and can be adapted for use on other social media platforms such as TikTok, Twitter, and LinkedIn. These platforms differ in their content formats, engagement metrics, and audience behaviors, yet the core multimodal approach can be extended to accommodate these variations.

As a suggestion for future works, one could consider combining methods beyond text and images, such as video content analysis or sentiment analysis of comments. Additionally, models could be explored that can predict not only initial engagement but also long-term brand recall and loyalty among audiences generated by influencer campaigns.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the readability and language of some parts of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Mohammad Javad Shayegan: Writing – review & editing, Supervision, Project administration, Conceptualization. **Nasim Kazem Shiroodi:** Writing – original draft, Visualization, Validation, Software, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

In this research, we utilized the Brand-Micro Influencer dataset from [6] (in the paper)

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